- €

•, ƒ „ - (IT & CS) … † „‡

(†) … ˆ ‰ Š ‹ Œ• ŒŒ

Ž•-4 Œ• ‘ ’ • • „ ‘

(Mini Processor Datapath Implementation using Logisim Evolution)

“” 1447 / • 2026	: •
–Œ– • ˆ | ˜ ™	:š – ‰ †›
„ ƒŒ œ / Œ œ •	: ž ŒŸ› Ž

.1 Œ Œ• ‹

.(Decoder) (Control Unit) (Registers) ﺪ • ‚ ƒ „ …†‡ ˆ ‰ Š‹Œ (ALU) •Œ‡ Ž † • ﺪ .Logisim Evolution • ‘ ƒ ’ “ ” -4 ‚ ‘ • ’ (Mini Processor Datapath) –† — ˜Š • ’ ‚ ‘ Š ™‡ ƒ š › Ž‹“ ﺪŠ ’ œ • ’ ž • ŸŒ ˆ ‰ ¡ž ¢ £ﺪ

.2 ¤‰ — • ¢ ™ ¡ •

.(AND, OR, XOR, ADD, SUB) “ “„ Œ‡ ‚ • ’ ¡ ‡ :(ALU) •Œ‡ Ž ¤

.(Clock) ¥Ÿ‘ ‡ •¦ ‡ ﺪ • Result Register ‚ š › ﺪ • Register B Register A § ‰ :(Registers) ‚ š ¨ ¤

.ALU ﺪ ‚ š ¨ § –‰ (Control Signals) ˜œ ‚ Š ª (” -3 • ’) Opcode ª §ﺪ‹ :(Decoder) (Control Unit) ¤

.3 ¤‰ — € £ ŒŠ •

3.1 ŒŠ ‰ ¦ ¥ (Full Adder)

:–ž § ¨• “ “„ Œ‡ ‚ «œ ’

Sum = A ⊕ B ⊕ Cin

Cout = (A · B) + (Cin · (A ⊕ B))

Cout	Sum	Cin	B	A
0	0	0	0	0
0	1	0	0	1
0	1	0	1	0
1	0	0	1	1
0	1	1	0	0
1	0	1	0	1
1	0	1	1	0
1	1	1	1	1

Ÿ› • Opcode ‰ Œ § £ € .4

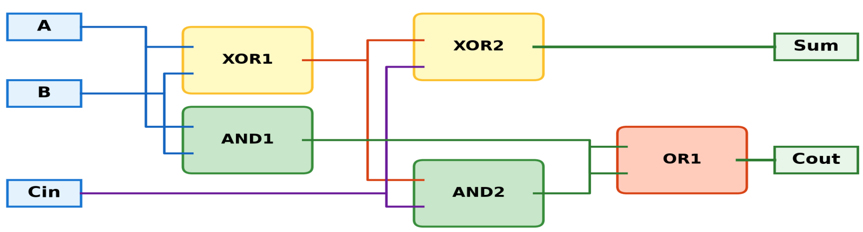
:¬ Œ Š ª ꞓ • (Multiplexer) - › ‡ – ª Ÿ • (Decoder 3:8) Ÿ £ ® Opcode ª §Ÿ

¨ © ªŒŸ	Sub ªŒ– Ÿ›	RegWrite •	ALU_Sel ‰	(Operation)	Opcode ‰
” § ” Œ‡ AND •	0	1	000	AND	000
” § ” Œ‡ OR •	0	1	001	OR	001
(§¯ °) Œ‡ XOR •	0	1	010	XOR	010
(A + B) ” -4 –¦‡ˉ – ®	0	1	011	ADD	011
(A - B) ‡ˉ‡ – ¬ ²	1	1	100	SUB	100
(RESERVED) ° • ³	0	0	101	NOP	101
(RESERVED) ° • ³	0	0	110	NOP	110
(RESERVED) ° • ³	0	0	111	NOP	111

†Œ™ ˆ Logisim Evolution ‘ —« .5

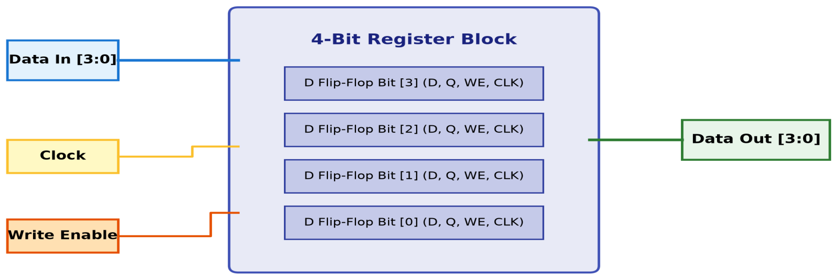
(Full Adder Subcircuit) ¥ ¦ ¬–« 5.1

Full Adder Circuit Schema (دائرة الجمع الكامل)

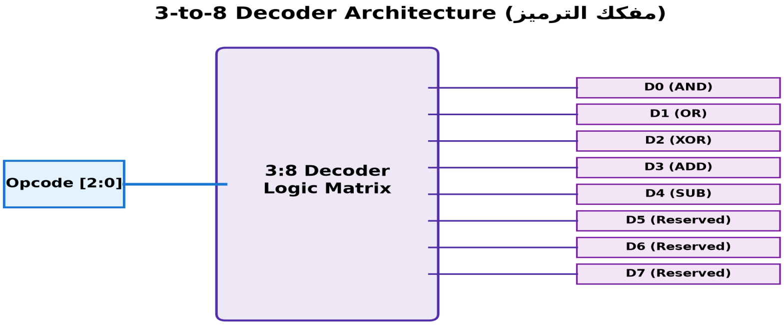


(Registers Block) - „ ˆ ‘ 5.2

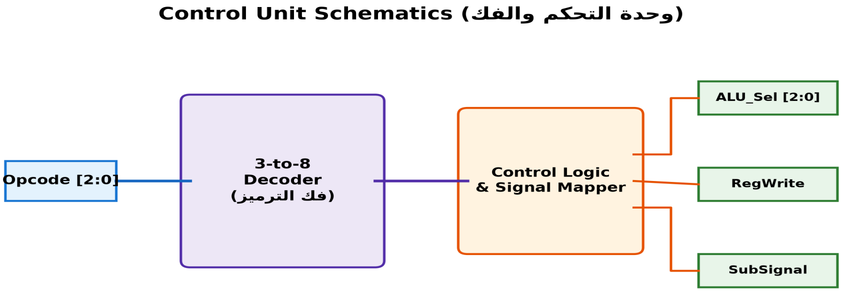
4-Bit Register Architecture (تصميم المسجلات)



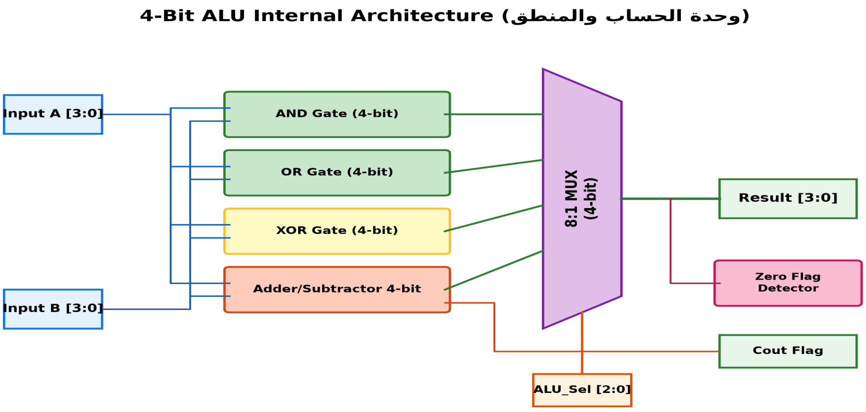
(Decoder 3:8 Subcircuit) ® ₠ ¯ ™ ‘ 5.3



(Control Unit Subcircuit) ˆ ‘ 5.4

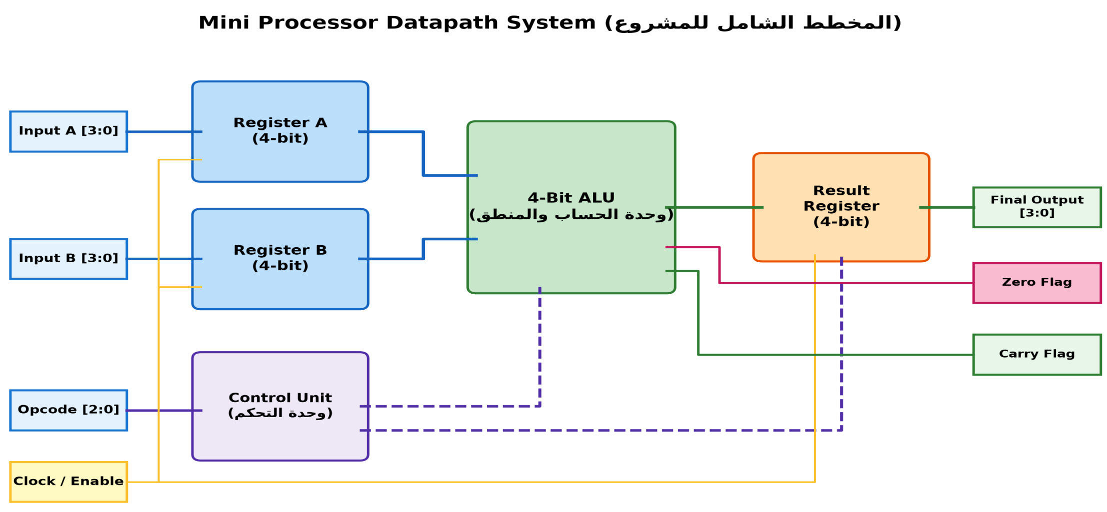


(ALU 4-Bit Subcircuit) °– … „ ˆ ‘ 5.5



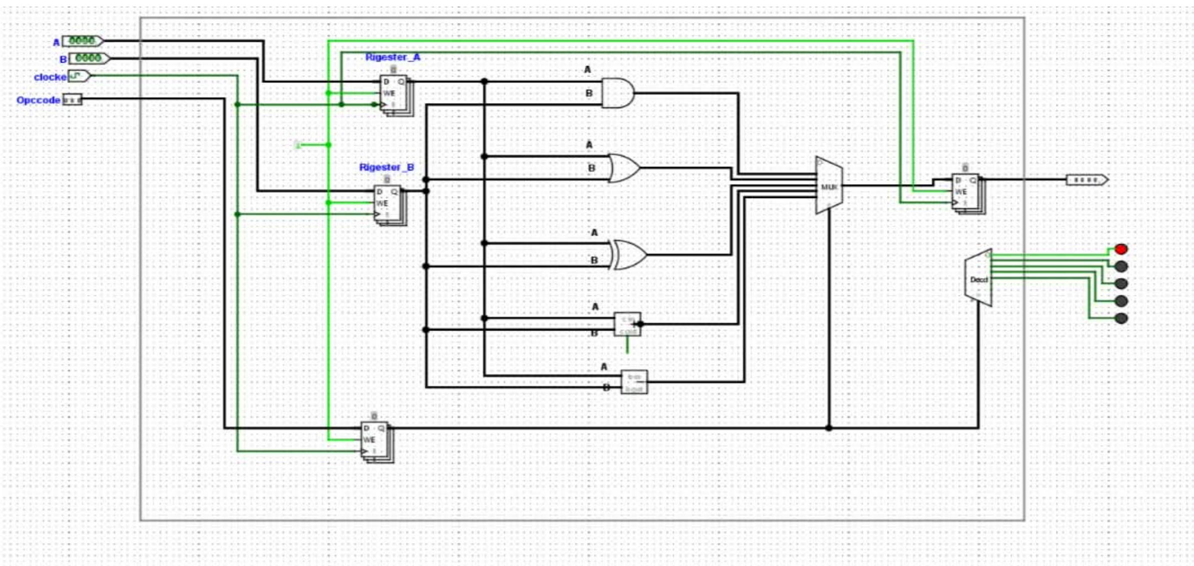
(Mini Processor Datapath) • „ ¥ • ¬–« 5.6

: ‚ ‘‹ ® • ´ –• £ µŠ ’ •Œ• • ’‹



(Œ « • ±™) Logisim Evolution ’ •Œ• ‹ Œ• ° – 5.7

:LED ‚ Ÿ ALU ‹ ‚ š¨ § ¶‹ · ‹ Logisim Evolution • ‘ • ˆ ‰ • § ª Ÿ – ŸŒ †
§¸ Š‹†



(Execution Cycle) Œ • ± ™ ‰ ¥ •• –² .6

.Register B Register A – £‡ꞓ • ¥Ÿ‘ § ’ Input B Input A , š › •Ÿ :(Fetch & Load) , ‘ Ÿ § .1

.RegWrite § ’ - › ‡ ¹®‹ Ÿ“ ‡ Š ª Decoder Š † (¨• 011 §¸) Opcode œ‹´ § › ꞓ:(Decode) „ ª .2

.Carry Flags Zero , Š ª ¨ ‡ MUX Š •ꞓ Œ‡ ± • ’ ALU µ ¨ :(ALU (Execute ¡ ‡ .3

. ¦‡¸ ª‰ Ÿ £ ’ Result Register – ¦£‡ ¨ ‡ ꞓ • • ¥Ÿ‡ °‹ :(Write Back) –¦£‡ ꞓ • .4

³ ¤ « .7

. ¸ꞓ , ¨ ’ ꞓŠ ’ • ®º‹ ‘ º ꞓ , ¥Ÿ‘ – ¸ • Opcode ƒ ® – œ Œ‡

„ ¨ ˜ ° »‡ ¨ “ Logisim Evolution • ‘ • ŽŠ ¨ ” Ÿ± .° ¬ ¨‡ ” -4 -† –¨ ’ , ‘ Ÿ Š ŠŸ › ™‡

¹ ‹ ¼ §¥